Olivia Tolliver

(404) 803-6880 | ot64@cornell.edu | Ithaca, NY | linkedin.com/in/oliviatolliver

Education

Cornell University, College of Engineering Ithaca, NY
Bachelor of Science in Mechanical Engineering, Robotics Minor (Systems and Design Specialization) *Expected May 2027*

Skills

CAD: SolidWorks, Fusion 360, Inventor, Onshape, GD&T, Technical Drawings, Tolerance Analysis, DFMA
Simulation & Analysis: ANSYS, FEA, MATLAB, Python, LabVIEW, GitHub
Hardware & Manufacturing: Manual Lathe/Mill, 3D Printing (FDM/SLA), Assembly, Soldering, Arduino, Raspberry Pi

Experience

Cornell Mars Rover Project Team Ithaca, NY
AstroTech Mechanical Subteam Lead *Feb 2025 – Present*

- Engineered an auger-based subsurface soil collection system for the University Rover Competition, replacing the previous scoop design to achieve 80% deeper access and 300–500% more sample volume per cycle.
- Designed the vertical translation assembly using an 18-in non-captive lead screw, dual 0.5-in hardened steel guide rods, and linear ball bearings to improve stability, reduce applied moments, and meet the 5 kg weight limit.
- Selected a Maxon EC-Flat 45 (70 W) motor with a 50:1 gearbox to deliver 5.7 Nm torque; experimentally demonstrated 150–300 N vertical load capacity, supporting a 110 lb rover under stall.
- Leads a growing 4-person mechanical subteam across 6 rover projects, defining project concepts, assigning project ownership to team members, managing the subteam budget, and contributing directly to design and development.

Cornell HAPPI Lab (Human Augmentation in Physical and Perceptual Interactions) Ithaca, NY
Undergraduate Researcher – Joint Pneumatic-Electromechanical Harvesting System *Mar 2025 – Jul 2025*

- Designed, fabricated, and tested 3 pneumatic pouch geometries for a wearable energy-harvesting system, using DAQ and MATLAB to identify the highest-performing design at 11.9 W, 17% above alternative configurations.
- Developed a foam insole mechanism converting foot-strike pressure into airflow to drive a 2-in. axial-flux generator; projected 86.9 mW electrical output, enough to power multiple low-power wearable EKG sensors.

MAE 2250: Introduction to Mechanical Design, Cornell University Ithaca, NY
Teaching Assistant *Jan 2026 – May 2026*

- Facilitated weekly lab sections in a design studio, mentoring student teams through hands-on prototyping, iterative design, fabrication, safe operation of shop equipment, and technical troubleshooting while fostering teamwork.
- Supported open-ended, multi-disciplinary design projects with Cornell CALS, providing technical guidance through concept development, testing, and refinement of mechanical solutions to meet functional requirements.

Projects

Drop-Down Vertical Storage Mechanism
Mechanical Design Engineer *Feb 2025 – Jul 2025*

- Identified a market gap and conceptualized a drop-down shelf for accessible vertical storage in small apartments.
- Led iterative design in Fusion 360 of a rolling-rail, magnetic-latch mechanism validated to a 15 lb load capacity; redesigned hinge and pin features after DFA testing to eliminate misassembly, improving build reliability.

Foundation Makeup Machine
Mechanical Product Designer *August 2026 – Present*

- Designing a mechatronic device that creates custom-mixed foundation by matching skin tone via camera-based color sensing, addressing inconsistent shade matches from uneven complexion and limited shade ranges in retail systems.
- Engineering the system's pigment cartridge, automated dispensing, and mixing mechanisms, using SolidWorks, DFMA, and microcontroller-driven controls to bring personalized shade-matching into a consumer device.

Leadership

National Society of Black Engineers, Cornell University Ithaca, NY
Conference Planning Chair *Aug 2023 – Present*

- Coordinate conference logistics, budgeting, and professional development programming for 55 members.

Alpha Kappa Alpha Sorority, Inc., Mu Upsilon Chapter Ithaca, NY
Financial Secretary *Apr 2026 – Present*

- Manage fundraising, university/donor funding, and dues; collaborate with Treasurer on chapter budgeting.